

Venous Pathophysiology

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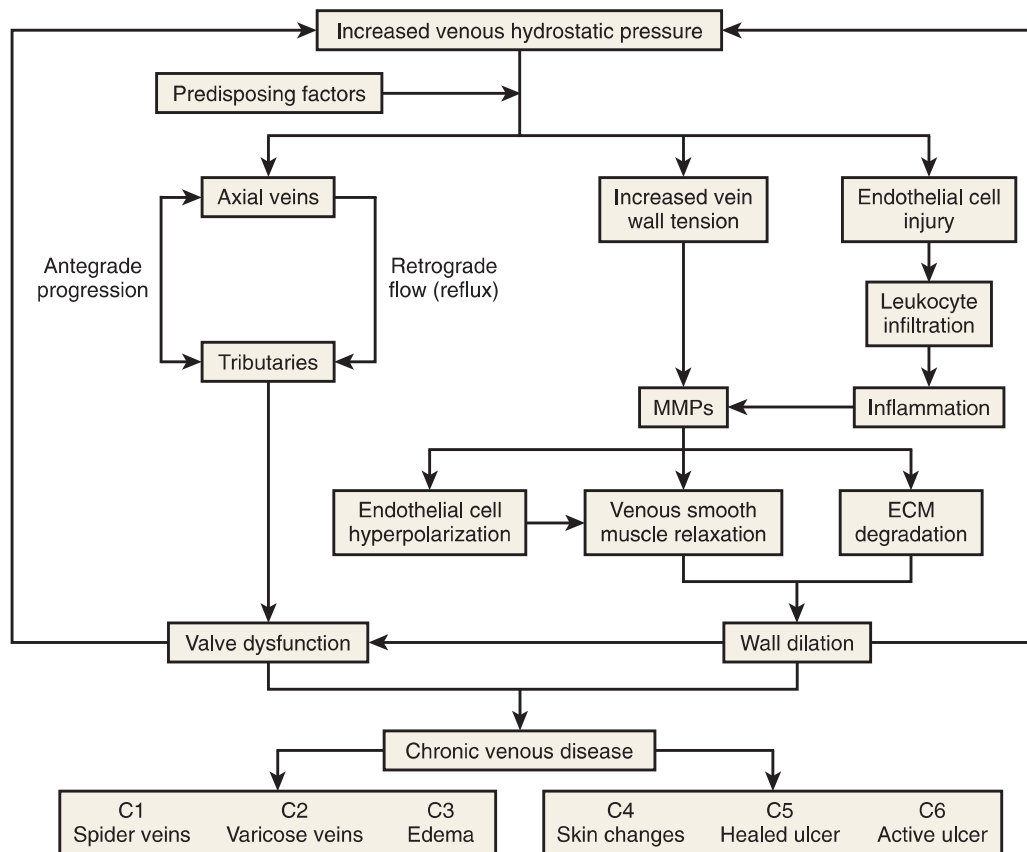
ETIOLOGY AND NATURAL HISTORY OF DISEASE

Primary Venous Disease

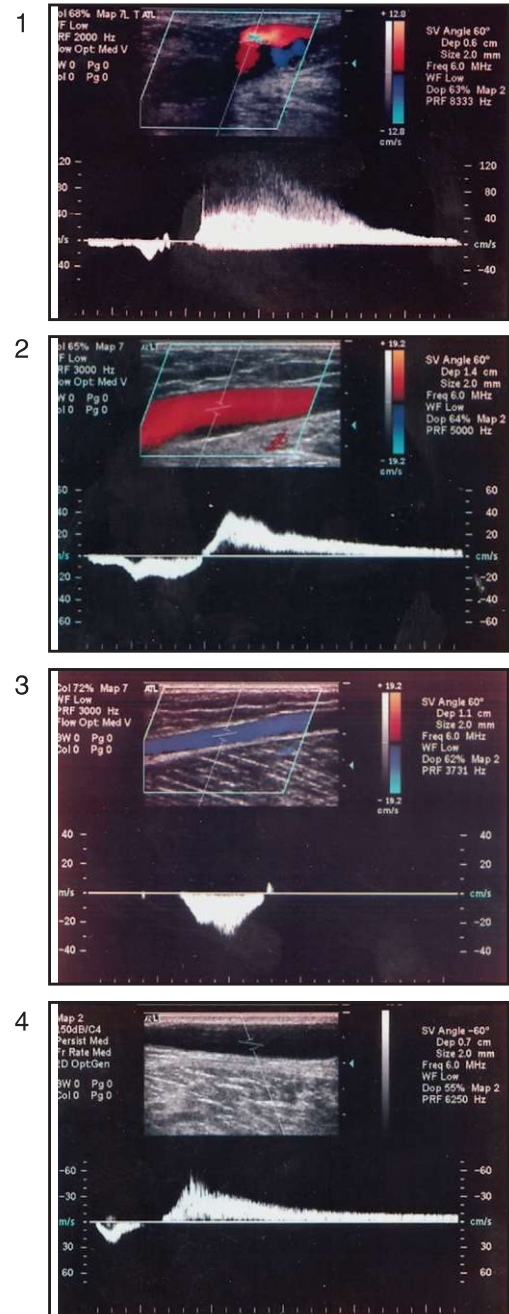
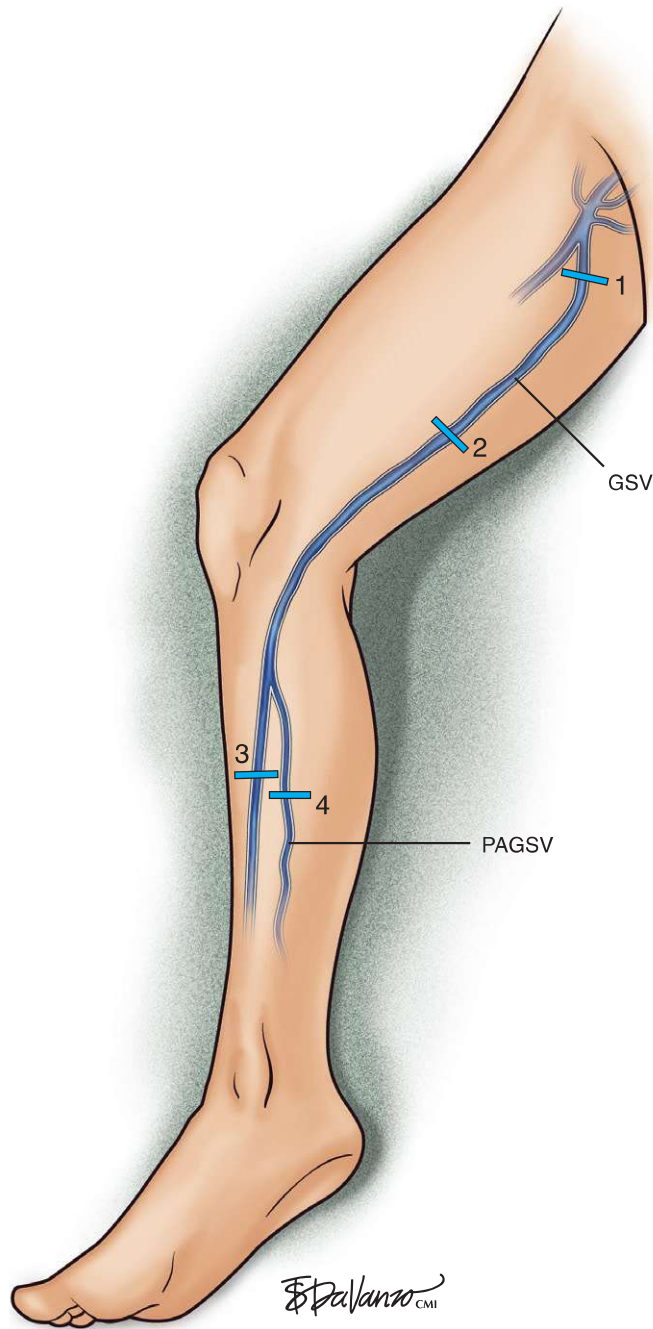
Primary venous disease affects two-thirds of patients with chronic venous disease (CVD). The most accepted theory is based on increased venous hydrostatic pressure transmitted to the vein wall, causing smooth muscle relaxation, endothelial damage, and extracellular matrix degradation with subsequent vein wall weakening and wall dilatation.¹ It has also been suggested that valve damage may occur because of local inflammation.² Leukocyte migration, plasma-granulocyte activation, and increased activity of metalloproteinases causing degradation of the valve leaflets support that theory.^{2,3} Figure 3-1 summarizes the pathophysiologic pathways of CVD.

Superficial veins are most commonly involved in primary CVD, followed by perforators and deep veins.⁴ It has been shown that reflux starts in superficial veins in more than 80% of the patients. In the early stages of CVD, reflux is found in the great saphenous vein (GSV) and its tributaries without almost any junctional involvement (Fig. 3-2). This is followed by reflux in the small saphenous vein (SSV) system (Fig. 3-3) and nonsaphenous veins (Fig. 3-4). Patients with competent saphenous, perforators, and deep veins may also present tributary reflux in 10%, with the GSV tributaries being affected in 65% of the cases.

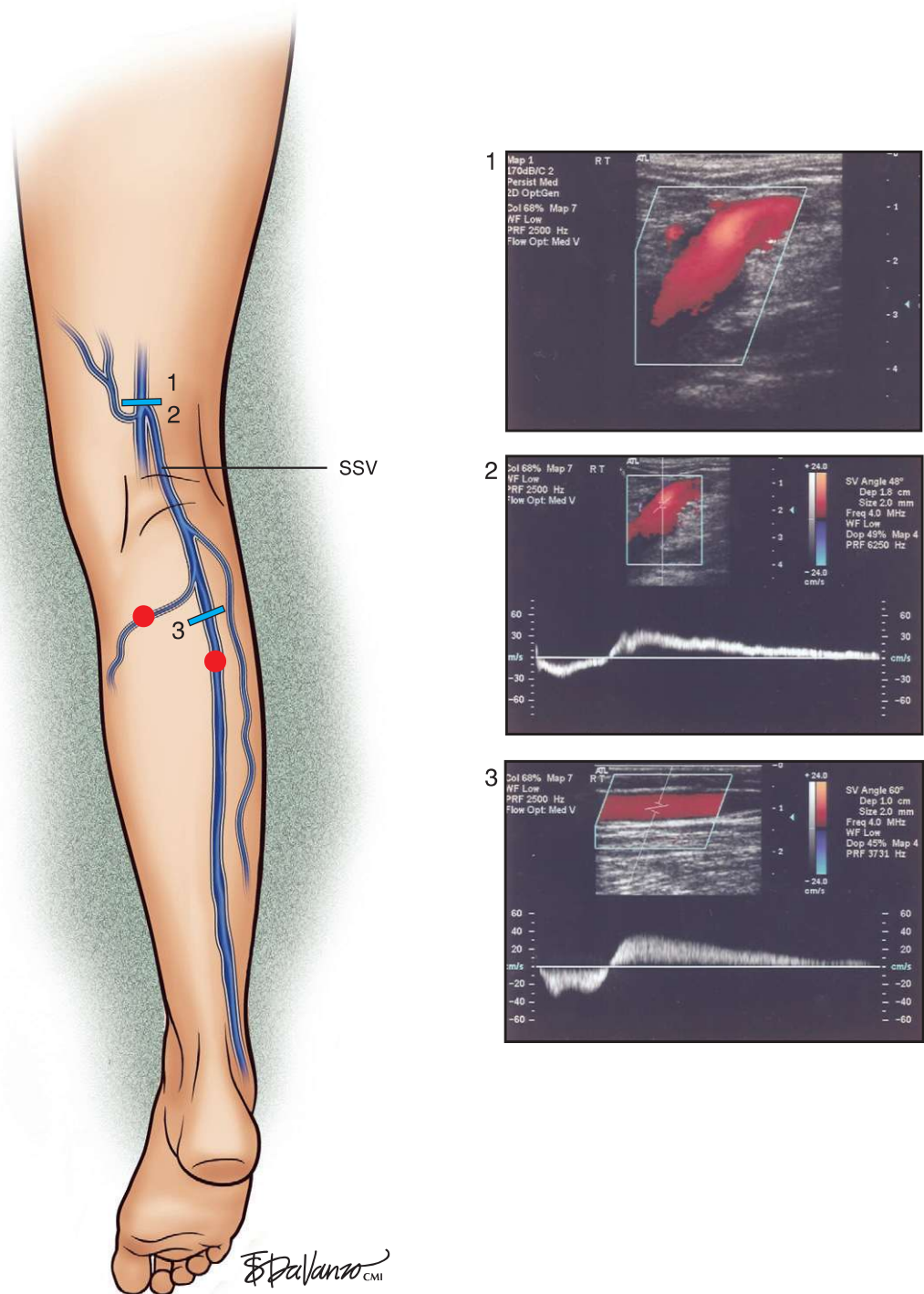
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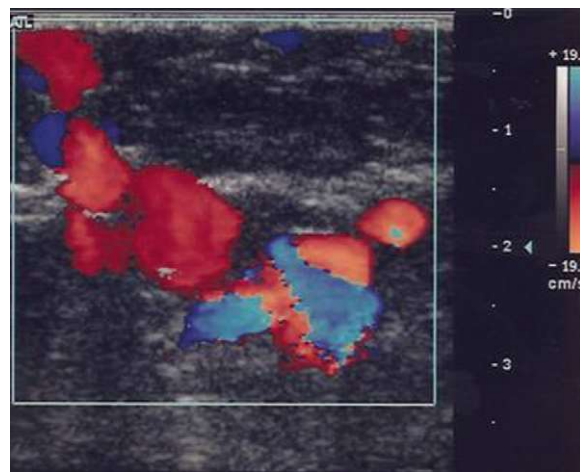
■ **Fig 3-1** Mechanisms of varicose vein formation. Increased hydrostatic pressure and wall tension in individuals with predisposing risk factors cause matrix metalloproteinase (MMPs) activation and changes in the endothelium and vascular smooth muscle function. In addition, leukocyte wall infiltration and inflammation activate MMPs and lead to extracellular matrix (ECM) degradation, venous wall weakening, and wall/valve fibrosis. Although a possible mechanism may involve primary valve insufficiency in both the axial and tributary veins, this likely represents a secondary event from primary venous wall changes and dilation. Persistent venous wall dilation and valvular dysfunction lead to increased hydrostatic pressure. MMP-mediated vein wall dilation with secondary valve dysfunction leads to chronic venous disease (CVD) and varicose vein formation. The early stages of CVD are maintained within the vasculature, leading to clinical signs of varicose veins, while more advanced CVD causes progression of chronic venous insufficiency affecting surrounding tissues and leading to skin changes and ulcer formation. (From Raffetto JD, Khalil RA. *Mechanisms of varicose vein formation: valve dysfunction and wall dilation. Phlebology* 2008;23:85-98.)



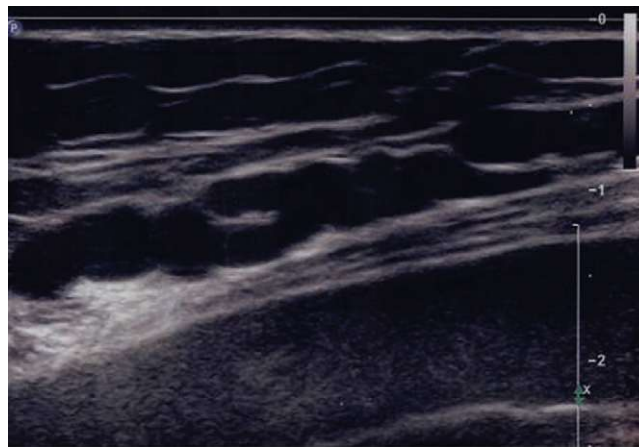
■ **Fig 3-2** Great saphenous vein (GSV) reflux from saphenofemoral junction to upper calf and the posterior accessory calf vein. The below knee segment of GSV is normal.



■ **Fig 3-3** Severe reflux in the small saphenous vein (SSV) in a patient who presented with CEAP classes 1 through 4, itching, and pain during prolonged standing. Reflux was also found in two tributaries and two perforator veins (red dots).



A



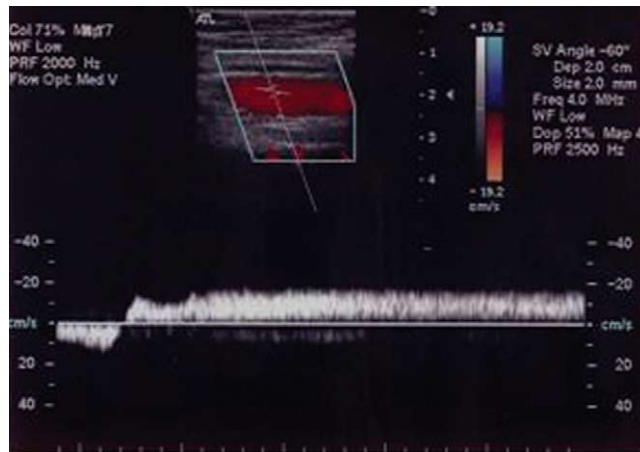
B

■ **Fig 3-4** Nonsaphenous vein reflux. **A**, Vulvar vein reflux in a female patient with a history of three pregnancies. The veins from this region are very tortuous and extend in a nonpredicted manner in the extremity. **B**, Sciatic nerve vein reflux giving rise to popliteal fossa varicosities that emerge in the posterolateral calf.

Isolated primary deep vein reflux is rare. It may present as either segmental or axial reflux extending from the femoral vein in the thigh to the below-knee popliteal vein. The most frequent location of primary deep vein reflux is the common femoral vein, followed by the femoral and popliteal veins⁵ (Fig. 3-5). Because most deep venous reflux is deemed to be caused by superficial venous reflux propagation, both common femoral and femoral vein reflux are associated with GSV incompetence and popliteal vein reflux is associated with SSV and/or gastrocnemial vein incompetence⁵ (Fig. 3-6). In addition, deep vein reflux has a shorter duration compared with superficial venous reflux. Association between deep and superficial venous reflux ranges from 5% to 38%.⁵⁻⁸

Perforator vein (PV) reflux in primary CVD always occurs in association with superficial vein reflux.⁹ Essentially, PVs become incompetent secondary to either ascending extension of superficial vein reflux or descending propagation of the reflux in a reentry fashion (Fig. 3-7). Most often, PV reflux originates from the GSV system and renders the deep veins incompetent in 13% of the cases.⁹ Deep vein reflux secondary to PV reflux is usually segmental and has a short duration.⁹

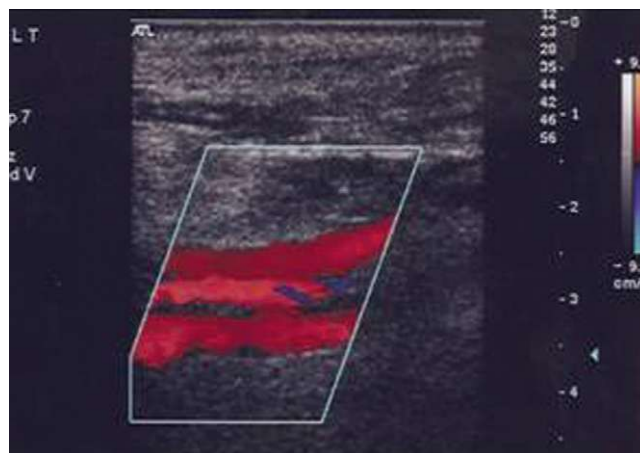
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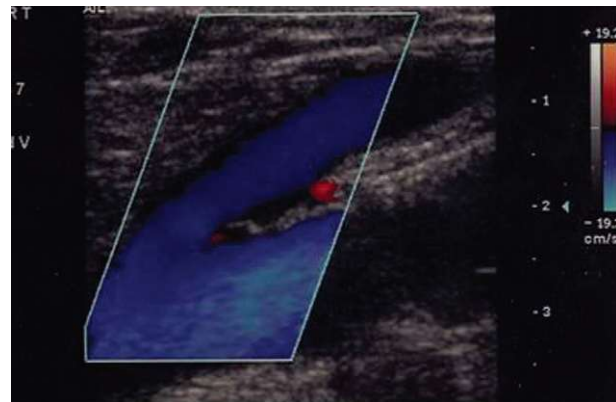


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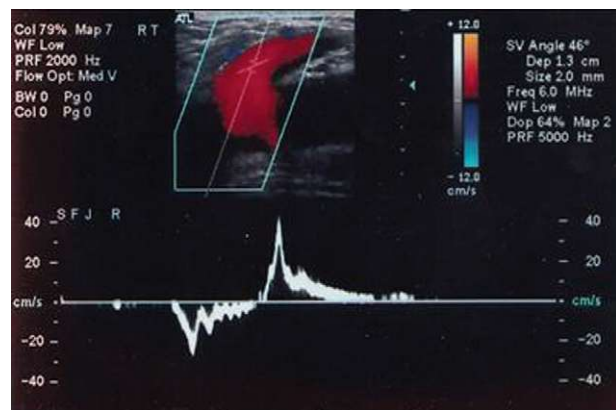


C

■ **Fig 3–5** Examples of postthrombotic reflux from three patients. **A**, Severe reflux in the femoral vein in a patient who had previous extensive deep venous thrombosis from the common femoral to calf veins. **B**, Popliteal vein reflux in a patient who presented with swelling and pain. **C**, Reflux in both posterior tibial veins, which have the same color as the artery in the middle.

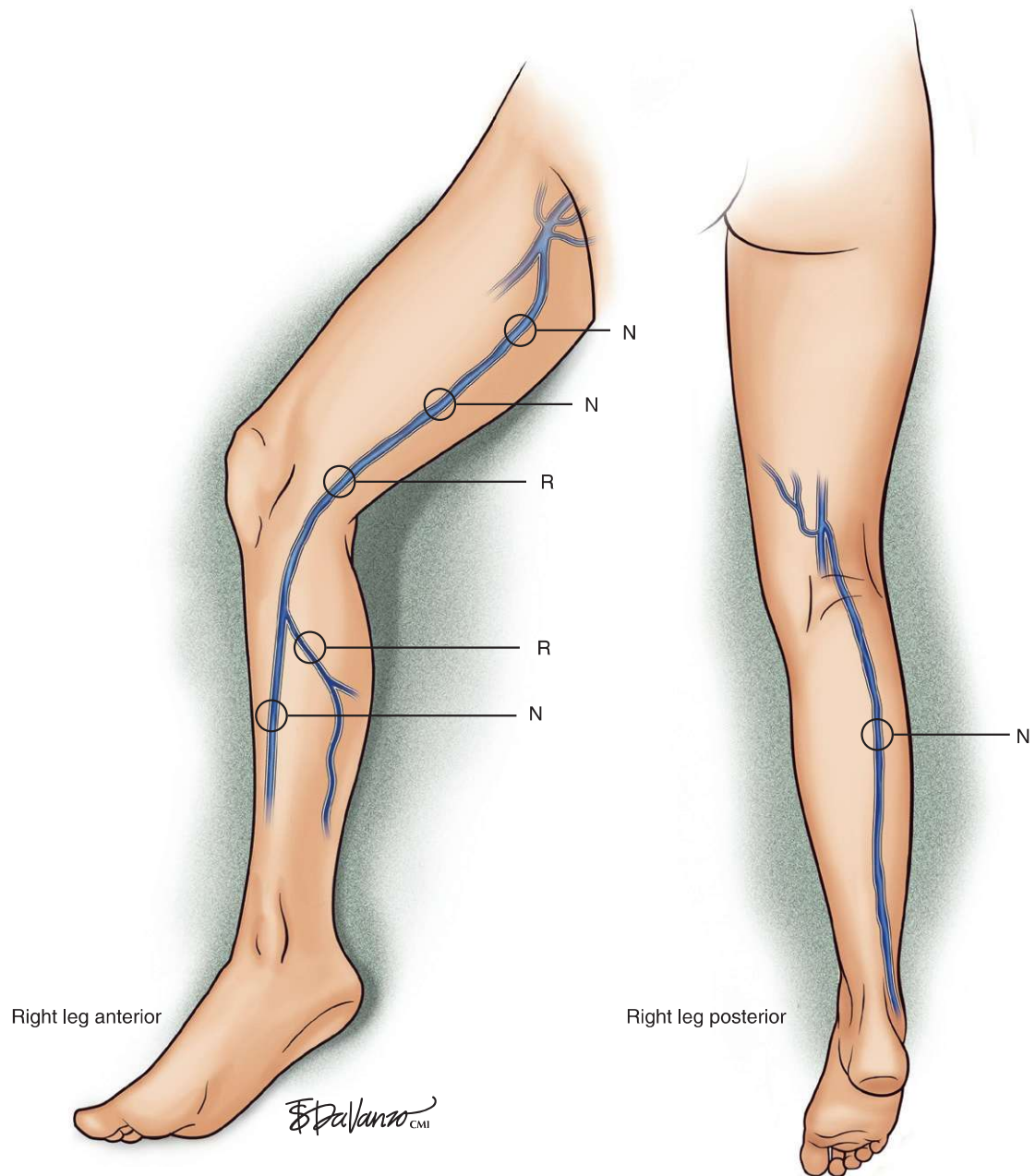


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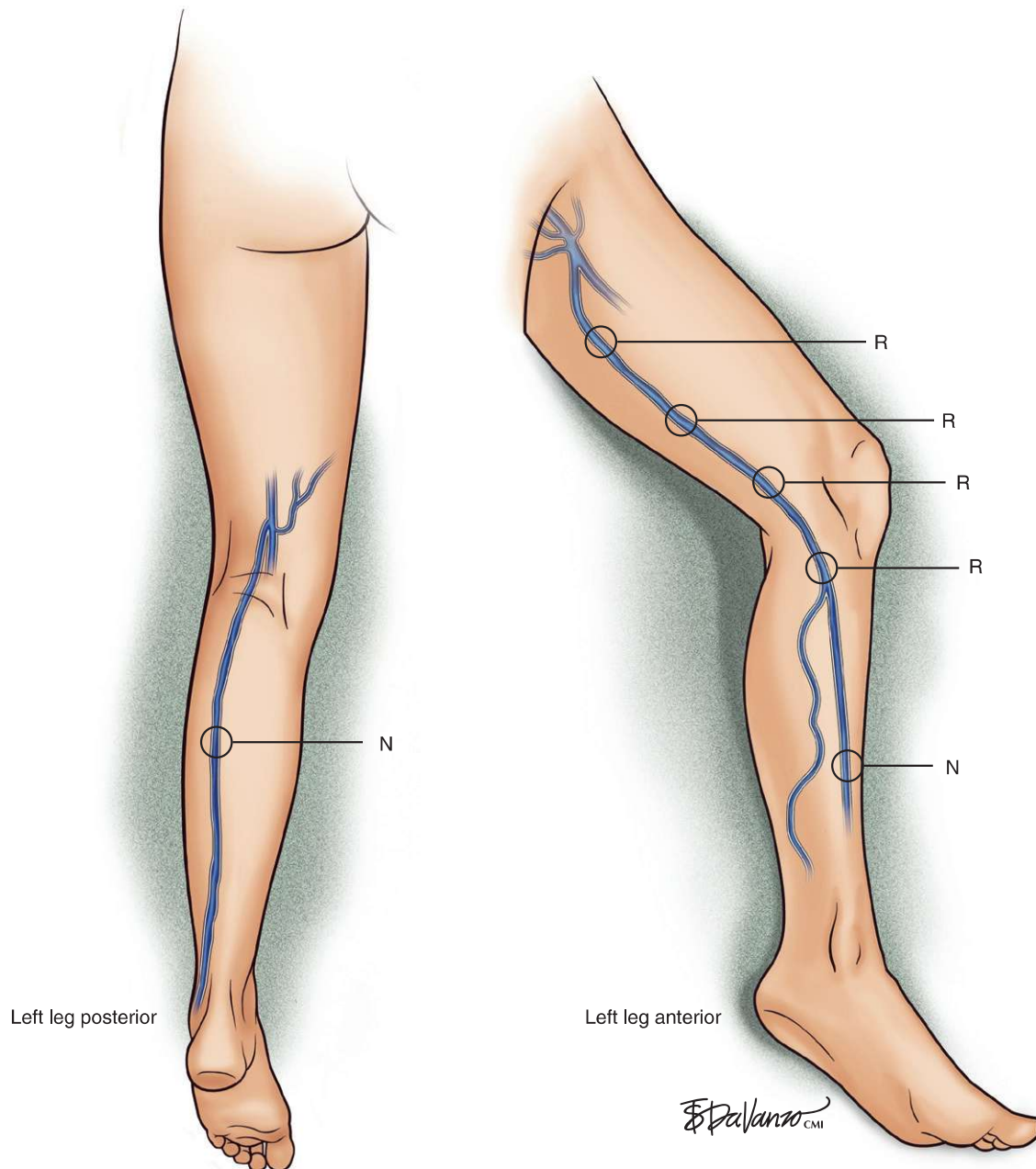
■ **Fig 3-6** Imaging of the saphenofemoral junction (SFJ). **A**, Cephalad flow is seen in the SFJ during distal compression. The saphenous vein has a normal diameter and is competent. **B**, SFJ reflux rendering the common femoral vein incompetent. In such cases in patients with primary chronic venous disease, correction of the saphenous reflux abolishes the common femoral vein reflux.



A

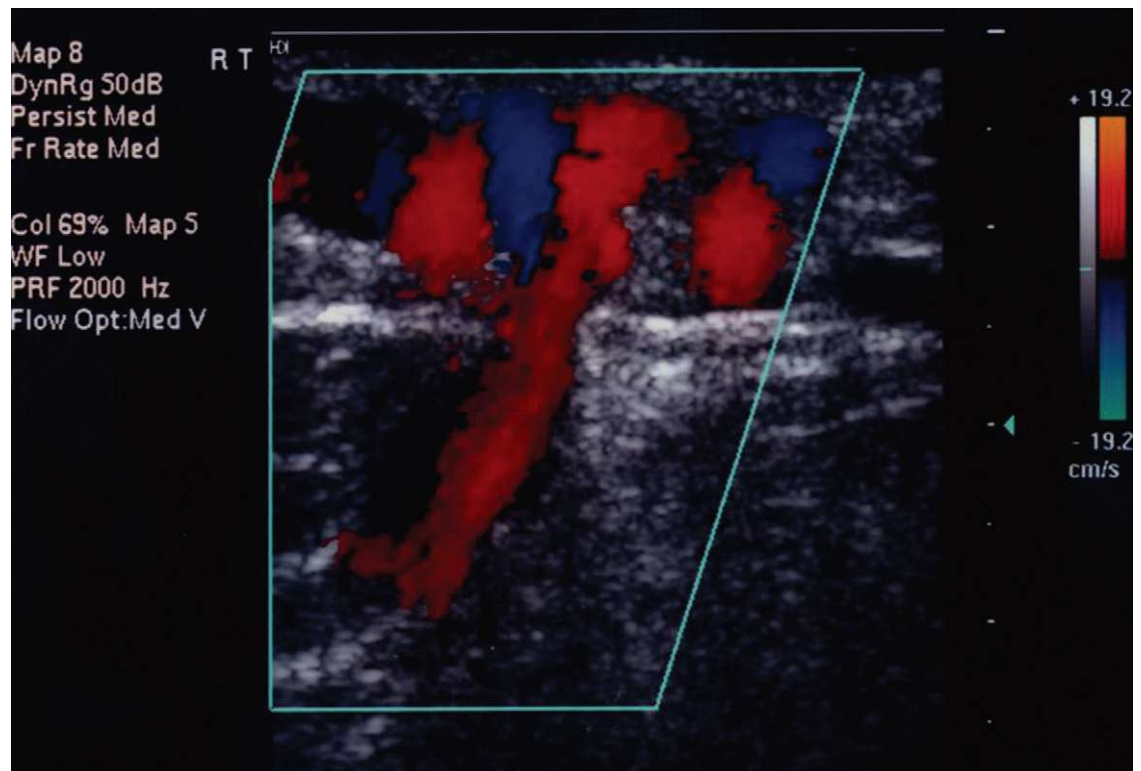
■ **Fig 3-7** Schematic drawing of the different patterns of perforator reflux development. **A** and **B**, Reflux developed in descending manner in a reentry perforator at the lower medial calf. *N*, Normal; *R*, reflux.

Continued



B

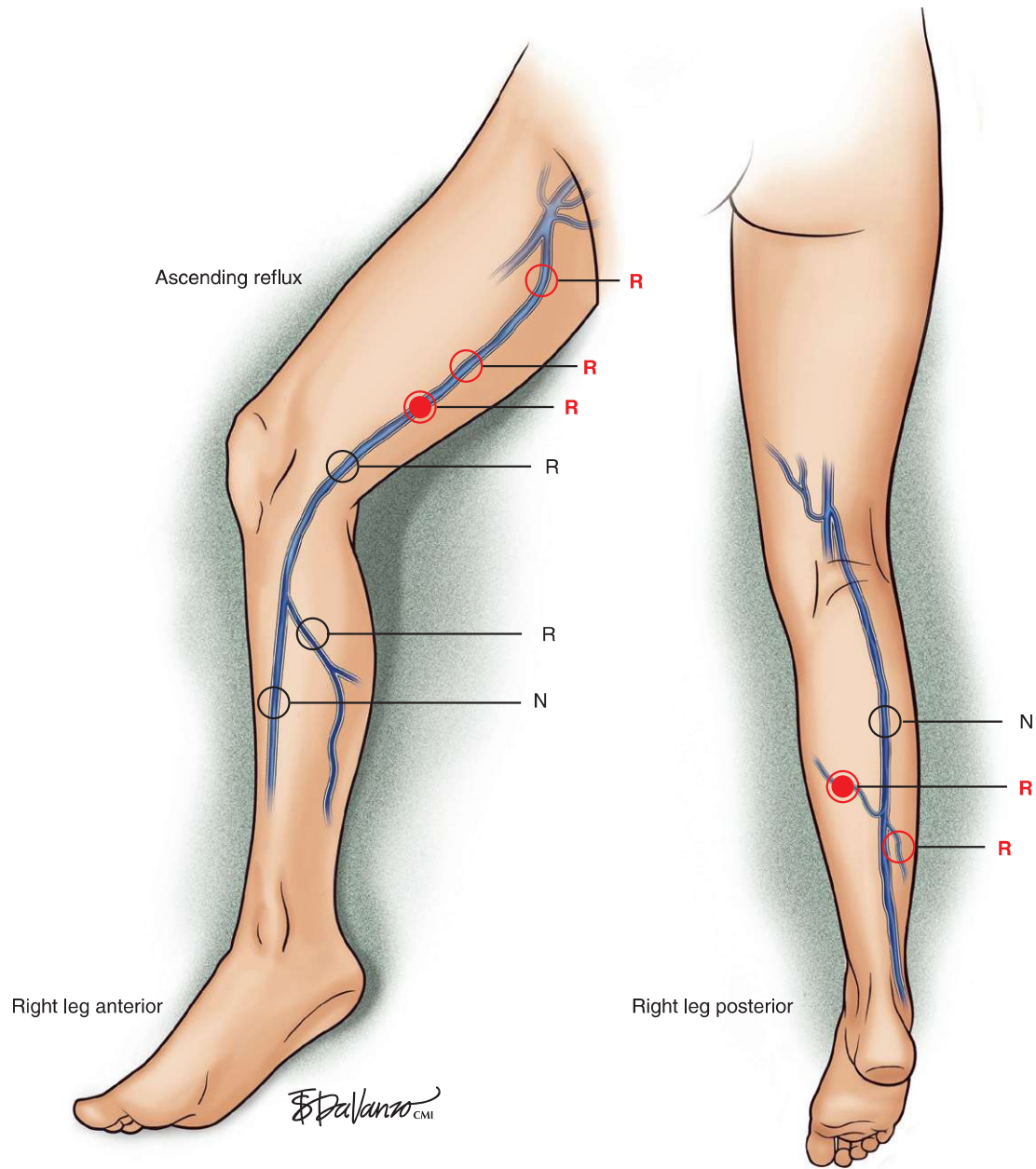
■ **Fig 3-7, cont'd** Schematic drawing of the different patterns of perforator reflux development. **A** and **B**, Reflux developed in descending manner in a reentry perforator at the lower medial calf. *N*, Normal; *R*, reflux.



C

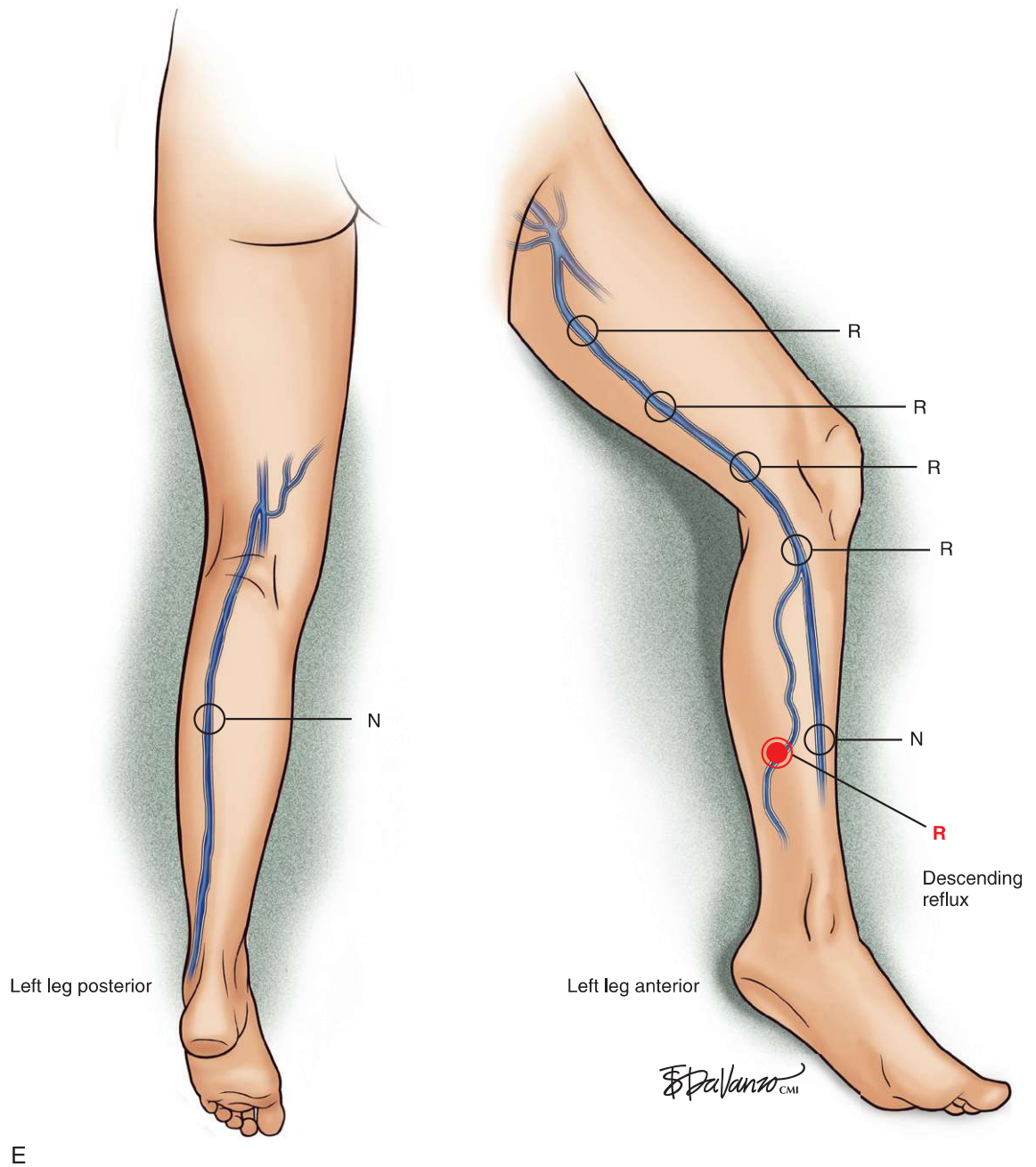
■ **Fig 3–7, cont’d C**, Baseline ultrasound examination in a 53-year-old female patient with a long-standing history of chronic venous disease. On the right limb, she had great saphenous vein (GSV) reflux from the lower thigh to the upper calf and in the posterior calf accessory vein. On the left limb, she had reflux in the GSV from the saphenofemoral junction to the upper calf and the posterior accessory calf vein. There was no perforator vein reflux at this time.

Continued



D

■ **Fig 3-7, cont'd D and E,** During the second examination at 38 months on the right limb, the patient developed ascending reflux in the GSV, a perforator vein at the thigh, a new reflux site in a posterior calf tributary, and a midcalf perforator. On the left limb, she developed a reentry type of reflux in a medial perforator vein from the posterior calf accessory vein. She had worsening of her disease from CEAP class 2 to 3 in the right limb and from class 3 to 4A in the left limb. She also became symptomatic in the left limb, with aching and itching along the varicosities of medial calf. *N*, Normal; *R*, reflux. The *red dot* indicates reflux in a perforator vein. The *red letters* and *lines* indicate the new sites of reflux in the second examination.



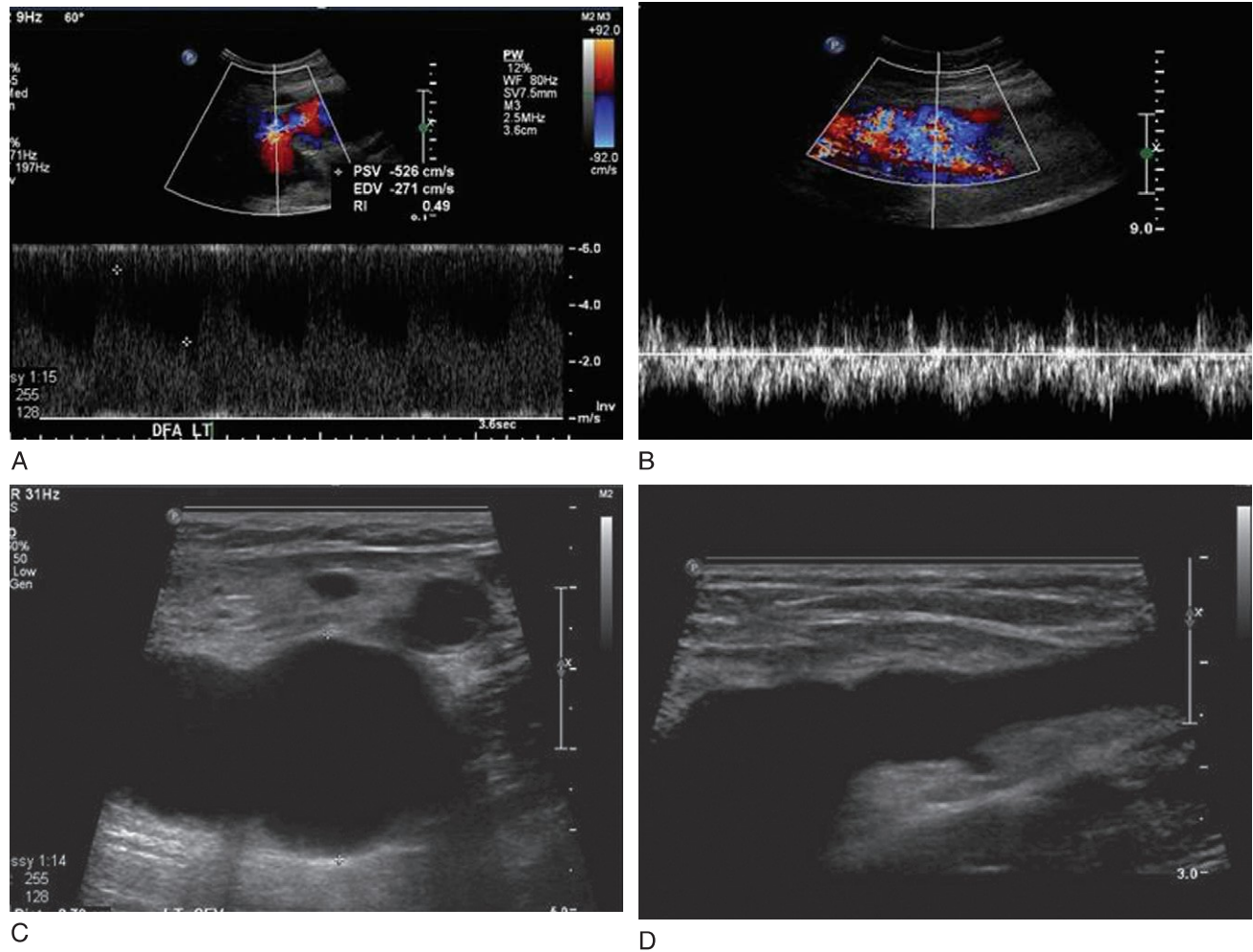
■ Fig 3-7, cont'd For legend see opposite page.

Secondary Venous Disease

Secondary venous disease is caused by a thrombotic event or is secondary to trauma. The incidence of arteriovenous fistula (AVFs) as cause of secondary CVD is reduced compared with postthrombotic CVD (Fig. 3-8).

Most frequently, AVFs are created when both common femoral vessels are inadvertently punctured during endovascular procedures or secondary to penetrating or blunt traumatic injuries. Animal models based on AVF creation have been used to explain the findings of chronic venous disease.¹⁰ Initially, lower resistance in the distal portion of the artery and pulsatile flow in the vein are noted. Venous hypertension supervenes, distending the veins and causing some degree of edema in the limb but no reflux. Subsequently, the valves are unable to appose what portends the reflux. Continuous venous hypertension and reflux entail valve atrophy and “arterialization” of the venous wall in the long term. In humans, the occurrence of CVD secondary to AVFs is rare and requires a long course to generate clinical impairment that frequently is devastating.

Deep venous thrombosis (DVT) is a result of stasis, endothelial damage, and/or hypercoagulable state. The predisposing factors for DVT and therefore secondary CVD are well established, including pregnancy, operations, immobilization, malignancy, trauma, and obesity. Patients who sustain inherited disorders (i.e., active protein C resistance, protein S and antithrombin III deficiency) are also prone to develop thromboembolic events.



■ **Fig 3-8** Arteriovenous fistula causing significant venous hypertension (swelling of the entire limb, pain, and discoloration) in a male patient after a gunshot wound in the left groin. **A**, Fistula between the deep femoral artery and the common femoral vein. There are high velocities with a low resistance pattern. **B**, Waveform with turbulent flow in the common femoral vein proximal to the fistula. **C**, The common femoral vein is dilated (2.8 cm) due to the local high pressure. **D**, Dilatation of the saphenofemoral junction and the great saphenous vein with continuous reflux (not shown).

Regardless of the initial thrombi formation, the majority of the limbs evolve with thrombus resolution. Notably, only one-third of the patients with secondary CVD develop postthrombotic syndrome (PTS), which consists of signs and symptoms such as pain, edema, heaviness, and intolerance to efforts that may progress to skin changes and ulcers (Fig. 3-9). Often, patients with PTS have a combination of reflux and obstruction (Fig. 3-10). PTS represents the most severe manifestation of secondary CVD, carrying significant socioeconomic impact, and is discussed in Chapter 15.

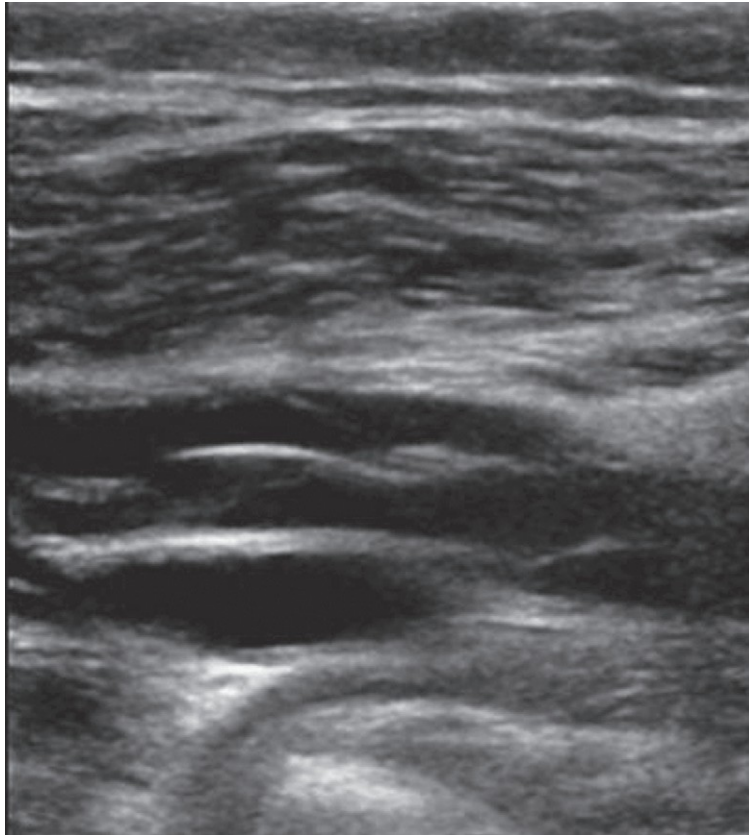
At the other extreme of secondary CVD are patients who develop DVT with no symptoms. This specific group of patients may have a total silent course of the disease for decades, being diagnosed with DVT during postoperative screening ultrasound or for nonrelated diagnostic purposes. The incidence of PTS in this subset of patients is low.

The location of the DVT and its extent have been investigated. Involvement of at least one proximal deep vein segment was proposed as a mandatory condition for the development of CVD. Three anatomic and hemodynamic patterns are found following an episode of DVT, reflux, obstruction, or a combination of reflux and obstruction. In that scenario, reflux is caused by destruction of the valves secondary to inflammation. It has been hypothesized that leukocyte infiltration in the venous wall causes amplification and activation of metalloproteinases, leading to venous wall and valve damage.¹

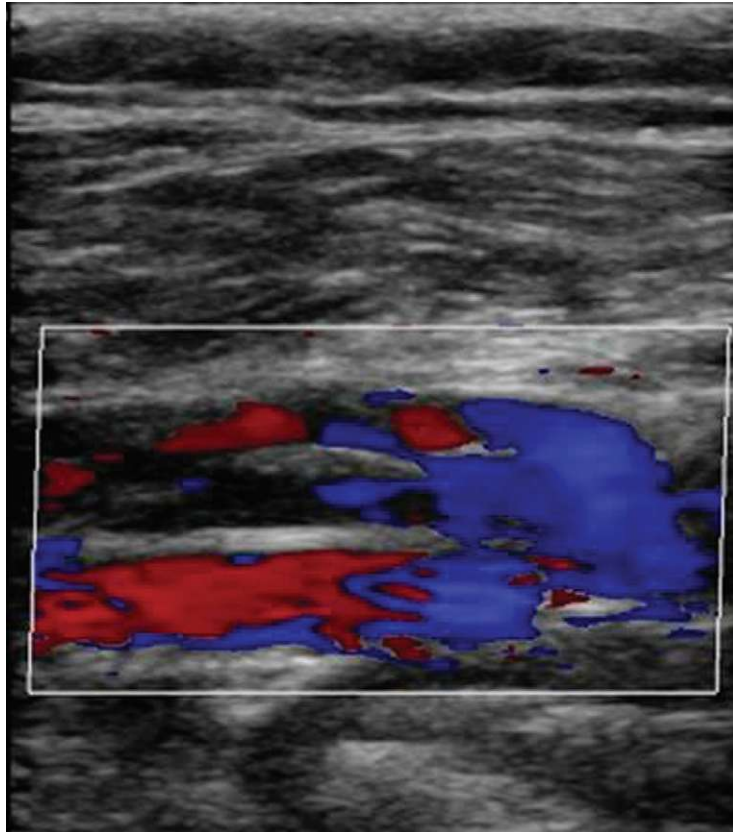
Failure in resolving an obstruction by either partial or complete recanalization of the thrombosed segment is found in less than 10% of cases.^{11,12} Partial recanalization is known to have higher incidence of reflux than complete recanalized segments, and calf veins are more prone to undergo recanalization than proximal segments.^{11,12} The role of location of the occlusion has



■ **Fig 3-9** Skin discoloration and swelling in patient with postthrombotic deep venous disease.



A



B

■ **Fig 3–10** Patient with leg edema and skin changes with partial recanalization of the popliteal vein. **A**, Synechiae are clearly seen in the lumen of the vein. **B**, Multiple flow channels are seen through the synechiae. Patients with reflux and obstruction are more likely to develop postthrombotic signs and symptoms.

been investigated. In a 1-year prospective study of 70 limbs, recanalization rate was lower in femoral veins, while all calf veins had complete recanalization. Finally, patients with reflux or obstruction present with milder changes and symptoms than those with both abnormalities. The presence of reflux and outflow obstruction in the same limb leads to higher rates of skin damage.^{13,14}

Congenital

Congenital vascular malformations contribute to 1% to 3% of CVD. Pure venous malformations are rare and present as an isolated cluster of veins that abuts surrounding tissues, including soft tissue and bone. The most common congenital malformation involving veins is Klippel-Trenaunay, which is characterized by varicose veins, limb hypertrophy, and port-wine stains¹⁵ (Fig. 3-11). Agenesis of valves and segments of the deep veins is rare but is known to cause CVD.¹⁶

Natural History

CVD slowly evolves over time and is classified based on its clinical (telangiectasias to skin damage), etiologic (primary, secondary, or congenital), anatomic (superficial, deep, or perforators), and pathophysiologic (reflux, obstruction, or both) patterns, forming the acronym CEAP.¹⁷ The clinical portion of the score is extensively used because of its simplicity, being cited as initial stages of CVD, including telangiectasias and varicose veins (C1-C2), and chronic venous insufficiency, including edema, skin changes, and ulcers (C3-C6).

Because the venous pressure increases in the lower extremity veins for hydrostatic reasons in during standing, it has been believed that reflux develops in a retrograde fashion. However, multiple studies have demonstrated that this is not true.¹⁸⁻²¹ Predominantly, reflux starts from the saphenous veins and their tributaries and progresses proximally, distally, or in both directions.²² In a longitudinal study of 116 limbs, progression of reflux occurred in 31. GSV and tributaries were the most common anatomic sites affected by reflux progression, followed by perforator veins. Seventeen limbs had extension of preexisting reflux in a proximal or distal direction, or both, and 14 limbs had reflux in a new segment that was independent of the preexisting site. Among patients with new signs or symptoms, documented reflux progression by duplex ultrasound was found in 53.8%, which was significantly higher than the 23.3% of patients without new symptoms ($p = .04$). Bernardini et al.²³ corroborated the findings that the reflux most frequently starts in the GSV and its tributaries, reporting progression of venous disease in 94% of the patients in a mean period of 4 years. Further evidence is provided by a recent study on wall characteristics²⁴ and interventional studies in which reflux in the saphenous vein was corrected after eliminating reflux in the saphenous tributaries.²⁵⁻²⁷

Secondary CVD was found to progress faster than primary CVD,¹² for reasons that are likely to be multivariate. The presence of reflux and obstruction aggravates the clinical status of patients, as shown in a study by Johnson et al.¹³ In a cohort of 64 patients, overall progression of the CEAP clinical class occurred in 31.5%. Notably, secondary CVD CEAP class 4 to 6 was noted in 4% of the limbs at a 1-year period, increasing to 25% at 5-year follow-up.¹⁴

Prandoni et al. analyzed the data of 1626 consecutive patients and found that residual thrombosis, unknown origin, and thrombophilia are risk factors for recurrent DVT over a period of 10 years.²⁸ Long-term follow-up with duplex ultrasound in a prospective cohort of 153 patients with recurrent DVT showed increased risk of skin damage (C4-C6) in patients with previous recurrent ipsilateral DVT.²⁹



■ **Fig 3-11** Klippel-Trenaunay syndrome in a young female patient who presented with pain and ulceration. The left limb has the typical discoloration and enlargement seen in patients with this syndrome. There were no anomalies seen in the deep axial veins. Reflux was found in multiple superficial, perforator, and muscular veins.



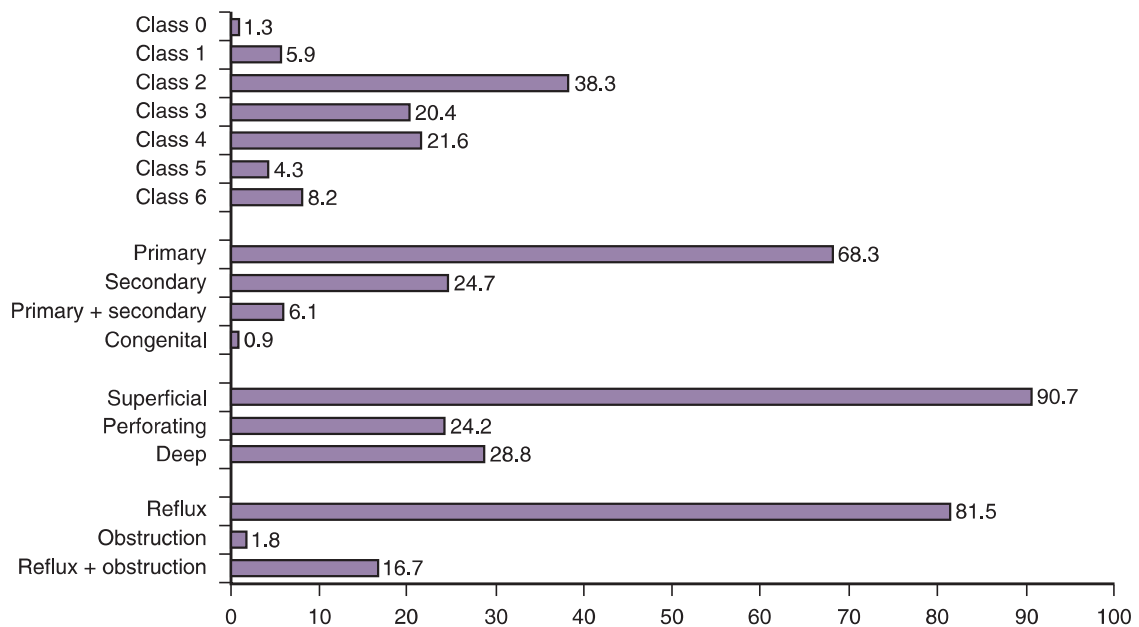
An overview of clinical distribution (reflux, obstruction, or both), classification, and pathophysiology of CVD in consecutive patients attending a vascular clinic is shown in [Figure 3-12](#). Most patients have primary vein reflux, which is most often found in the superficial veins and varicose veins. Skin damage is present in about one-third of the patients, while isolated deep vein reflux and obstruction are uncommon.³⁰

PEARLS AND PITFALLS

A complete understanding of venous pathophysiology is essential to offer an adequate treatment for patients with CVD. The relationship between deep and superficial venous reflux, obesity, and dilated veins with no reflux deserves attention.

Obesity

An association between obesity and CVD has been identified. The role of obesity as a causative versus an aggravating factor remains debatable. It is known that obese patients have higher intraabdominal pressure compared with nonobese patients and have reduced distensibility of the veins that could explain the higher incidence of CVD. Van Rij and associates³¹ reported an increased incidence of more severe CVD (CEAP 4-6) in obese patients than in nonobese patients. Interestingly, the authors concluded that obese patients have better venous calf muscle pump than nonobese patients. Sedentary behavior may explain the reduced effect of the pump to compensate reflux.



■ **Fig 3-12** Presentation of 1000 consecutive limbs with chronic venous disease (CVD) according to CEAP classification. In the anatomic classification, for simplicity only the overall contribution of each system is shown. This classification has allowed better communication and comparisons in the medical literature as chronic venous disease terminology is more specific and has been adopted worldwide.

Padberg and colleagues also reported a correlation between higher body mass index (BMI) and severity of the CVD³² in a cohort of morbidly obese patients (BMI >40 kg/m²). Obese patients had longer mean ulcer healing time, up to 7 months. Notably, 62% of those patients had no anatomic evidence of reflux despite severe CVD changes.³² Perhaps the disease in that subset of patients may be secondary to microcirculatory changes or even segmental venous hypertension associated with lymphatic drainage impairment. Therefore, a high level of suspicion should be raised when evaluating such patients because venous reflux may not be the cause of skin damage.

Effects of Superficial Vein Reflux on the Deep Veins

The role of superficial venous reflux as a causative factor for deep venous reflux has been investigated. Over the past three decades, selected patients with CVD who presented with deep venous reflux, advanced venous stasis, and skin damage have been subject to a multiple techniques to improve venous return, including valve transposition, repair, or even axillary vein transfer.³³

Because isolated deep venous reflux is rare, the current concept of flow overload in the superficial venous system causing venous hypertension and valve dysfunction has been stated. Transmission of the reflux through perforators and saphenofemoral junctions is likely to explain deep venous reflux in association with superficial venous reflux.

Several authors have advocated treatment of superficial venous reflux instead of direct procedures to correct deep venous reflux. Walsh et al.⁶ performed GSV stripping in 29 limbs with primary CVD, including only CEAP classes 1 through 3, and achieved success abolishing femoral vein reflux in 93% of the cases. In a similar subset of patients including only CEAP classes 1 through 3, Sales et al.⁸ achieved hemodynamic normalization of the deep veins in 94% of the patients. However, at the other extreme of the disease presentation, Padberg et al.³⁴ treated 11 limbs with active ulcers and obtained hemodynamic success in only 27%. All perforators were ligated and GSV stripping was performed. Nonetheless, the patients did improve, and the ulcers healed with no recurrence at mean 16-month follow-up. Ciostek et al. analyzed photoplethysmography and duplex ultrasound reflux parameters following GSV ablation in 11 patients with secondary CVI and found no statistically significant improvement in deep venous reflux.³⁵

The rationale for treating superficial venous reflux before any deep venous intervention is based on clinical and hemodynamic findings. Patients with superficial reflux associated with deep venous reflux present with milder deep vein impairment than do patients with isolated axial deep venous reflux, suggesting propagation and therefore response to a secondary mechanism as seen in study of 152 limbs.⁵ In addition, deep vein reflux is abolished in patients with initial disease after saphenous vein interruption, corroborating the former mechanism of superficial venous reflux propagation.^{6,8} Hemodynamic results may be compromised in patients with skin damage or secondary CVD due to prolonged inflammatory changes and more severe dilation of the deep veins.

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